



MyTrip

Budget Road Trip Planner

CS 4398 – Software Engineering Project

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Mustafa Al Azzawi · Angelica Alvarado · William Parker

Route



Cost



Budget





Project overview

One simple workflow for planning a road trip

1 Plan a route

Traveler enters starting point, optional waypoint, and destination.

2 Estimate costs

MyTrip calculates fuel cost, total estimated cost, and drive time.

3 Compare budget

The summary shows if the trip is within the traveler's target budget.

The screenshot shows a web browser window with the URL 'localhost'. The page has a navigation bar with links: Home, Plan Route, Route Options, Summary, Account, and Admin Portal. There are also buttons for 'US Gas Avg: \$3.42/gal' and 'Guest'. The main content area features a header 'AUTOMATED ROAD TRIP PLANNER' and a sub-header 'Effortless Trips. Direct Bookings.' followed by a description: 'Auto-computes gas prices, vehicle efficiency, distances, and offers direct booking for stays & attractions.' Below this is a form with two input fields: 'STARTING POINT' with the value 'Austin, TX' and 'DESTINATION' with the value 'San Antonio, TX'. A 'Generate Smart Route' button is at the bottom of the form.

Home page starts the planning flow with a clean route search.

- Web-based prototype
- Easy to use for travelers
- Designed for CS 4398 final project



Project purpose

The goal is to reduce scattered trip planning

Before MyTrip

- Maps on one website
- Hotels on another website
- Gas prices somewhere else
- Budget math done manually



With MyTrip

- Route display
- Cost estimation
- Budget comparison
- Saved trip plans

Main idea: help travelers understand the route and cost before committing to a trip.



Main features

Implemented features kept simple for the prototype



Route display

Google Maps shows the trip route.



Waypoint

Optional stop between start and destination.



Fuel cost

Mileage, MPG, and gas price estimate fuel cost.



Budget

Total cost is compared to the target budget.



Saved trips

Trips can be saved, reopened, edited, updated, and deleted.



Account filter

Saved trips are filtered by traveler email.



Website walkthrough

The interface moves through clear pages

1 · Plan Route

MyTrip | AUTO BUDGET | Home | Plan Route | Route Options | Summary | Account | Admin Portal | US Gas Avg: \$3.37/gal | Guest

1. Trip Parameters

STARTING LOCATION
Austin, TX

PIVOT-STOP / WAYPOINT (OPTIONAL)
New Braunfels, TX

DESTINATION
San Antonio, TX

VEHICLE TYPE (AUTO MPG PRESET)
Sedan 28 MPG | SUV 20 MPG | Hybrid 50 MPG | Truck 18 MPG

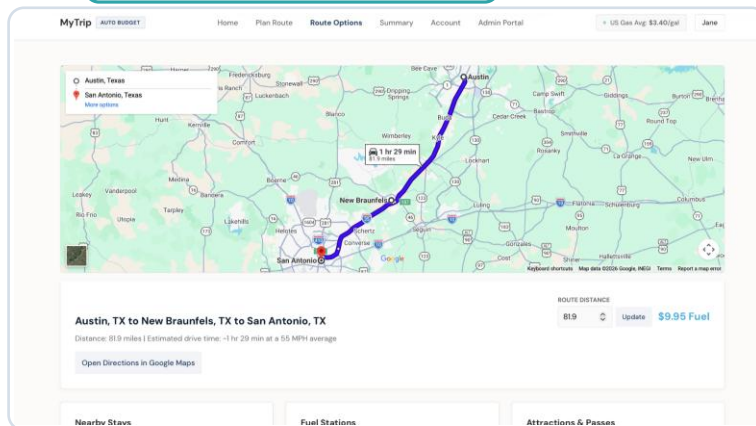
TOTAL TARGET BUDGET (\$)
200

Calculate Route & Cost Options

2. Saved Trip Plans

No saved trip plans yet.

2 · Route Options



3 · Summary

MyTrip | AUTO BUDGET | Home | Plan Route | Route Options | Summary | Account | Admin Portal | US Gas Avg: \$3.38/gal | TRAVELER

Trip Overview & Cost Breakdown

ROUTE DETAILS
Austin to New Braunfels, TX to San Antonio
Vehicle: Sedan (28 MPG) @ \$3.38/gal US Avg

Fuel Cost (Auto-Calculated)	\$9.89
Lodging	\$0.00
Activities & Tickets	\$0.00
Total Estimated Cost	\$9.89

TARGET BUDGET: \$200.00 | WITHIN BUDGET

Group Cost Splitter
Split total expenses among travelers
Travelers: 1 | Cost per person: \$9.89

Confirmed Bookings & Pass Vouchers

0 ACTIVE
No active reservations found.

Saved Trips in Account

No saved trip plans yet.

4 · Account

0 BUDGET | Home | Plan Route | Route Options | Summary | Account | Admin Portal | US Gas Avg: \$3.42/gal

TRAVELER

traveler@example.com
VERIFIED ACCOUNT
Log Out

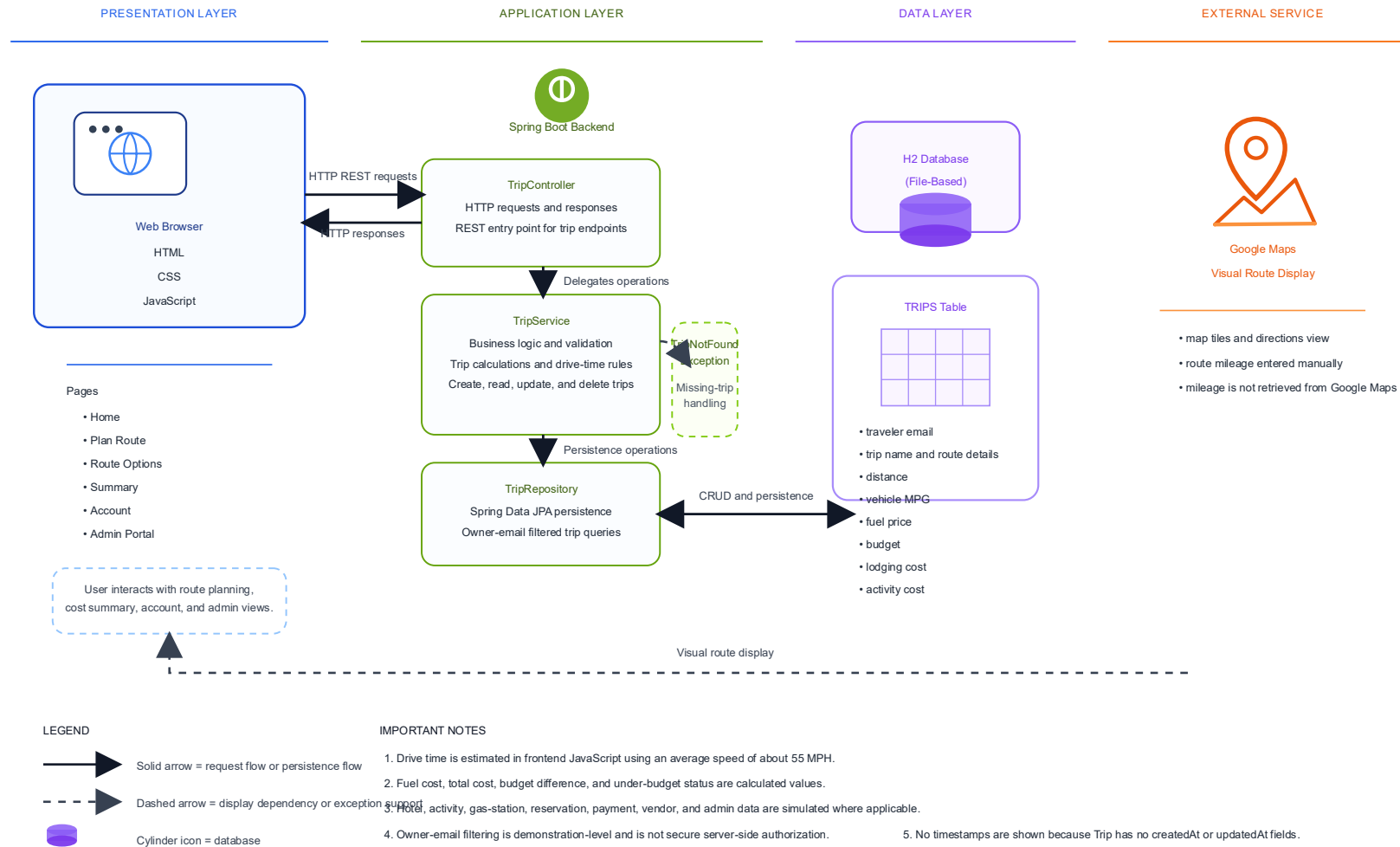
My Saved Trip Plans

Austin to New Braunfels, TX to San Antonio
Budget: \$200.00 • Estimated total: \$9.89 • Trip #2
Open Edit



System structure

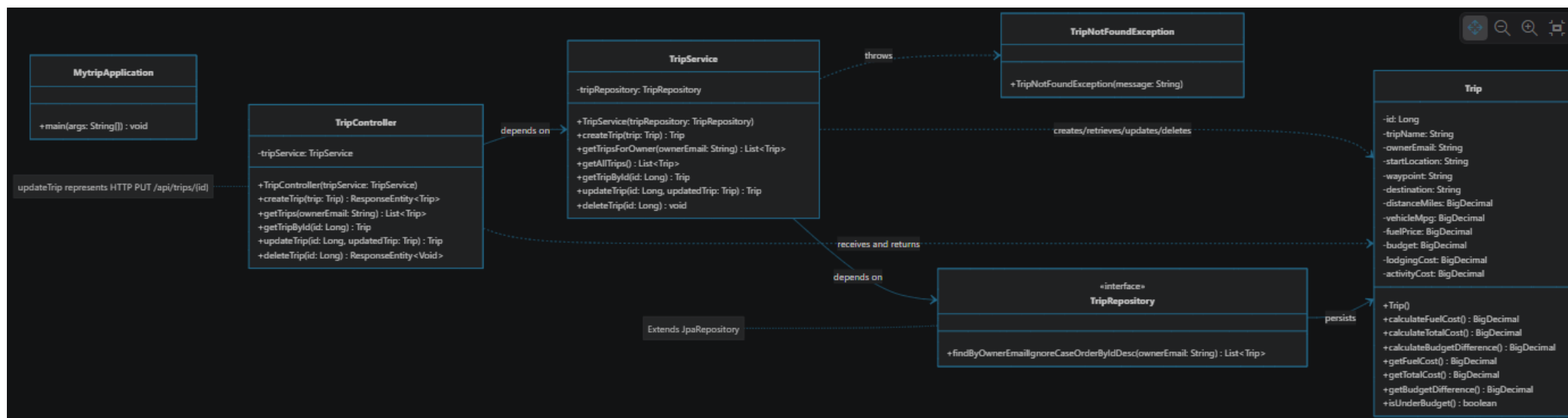
Layered design used by the application





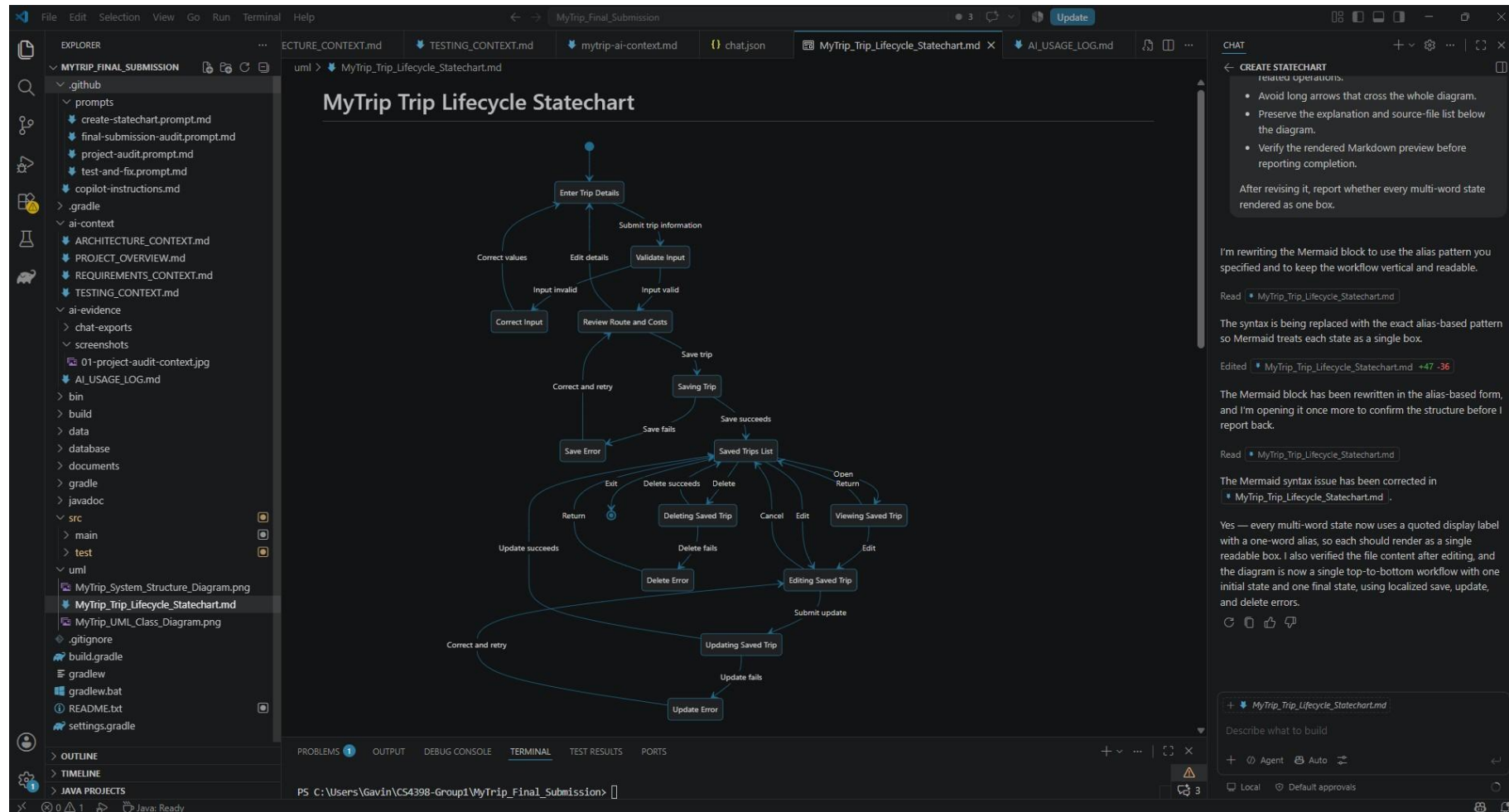
UML class diagram

Main classes and relationships



Lifecycle Statechart

Main classes and relationships





Algorithmms

Simple calculations create the trip estimate

Fuel Cost

$(\text{Route Distance} / \text{Vehicle MPG}) \times \text{Fuel Price Per Gallon}$

Total Estimated Cost

Fuel Cost + Lodging Cost + Activity Cost

Budget Difference

Budget – Total Estimated Cost

Tested example

- 81.9 miles
- 28 MPG vehicle preset
- \$3.38 per gallon
- $\approx$ \$9.89 fuel cost
- Drive time $\approx$ 1 hr 29 min at 55 MPH



Acceptance testing

The main traveler workflow was tested

Manual Acceptance Testing

10

TESTS EXECUTED

10

PASSED

0

FAILED

- Login and account access
- Create, calculate, save, reopen, and delete a trip
- Invalid mileage and invalid budget validation
- Logged-out save restriction
- Email-based trip filtering

Automated Acceptance Testing

29

TESTS EXECUTED

29

PASSED

0

FAILURES

0

ERRORS

0

SKIPPED



Known limitations

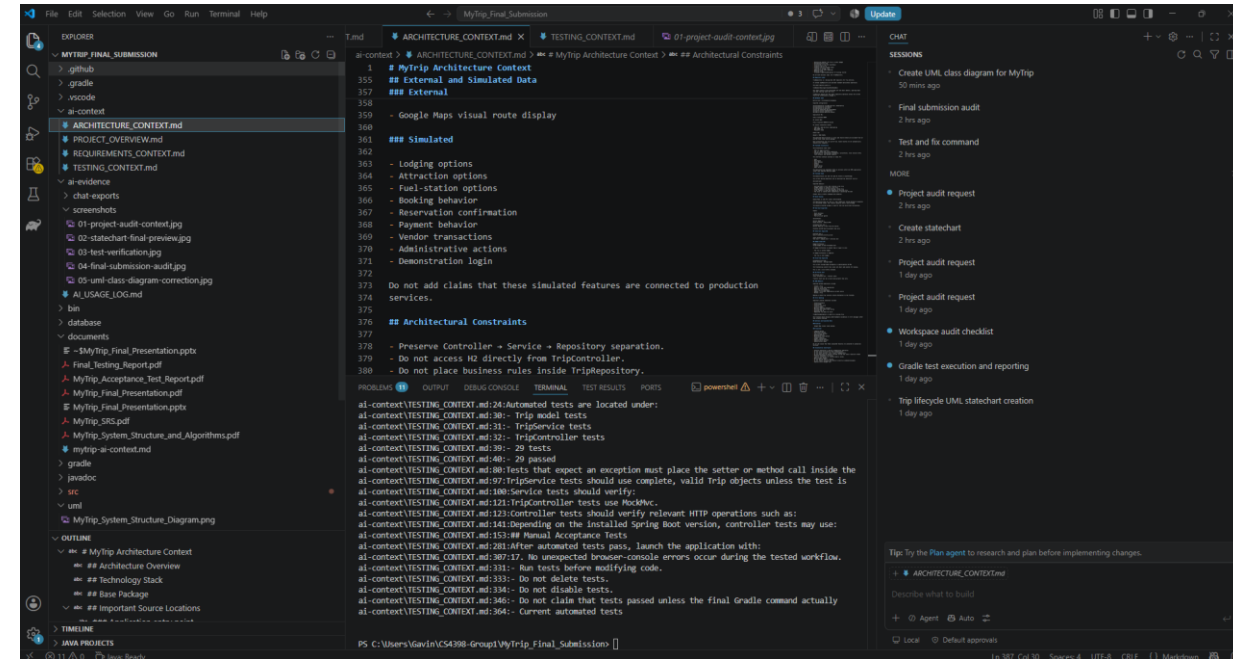
These are acceptable for the academic prototype

- Route mileage is entered manually
- Drive time uses an average-speed estimate
- Hotel, attraction, and gas-station data is simulated
- Reservations and payments are simulated
- Login and email filtering are demonstration-level, not secure authorization
- The H2 database is local to each computer

IDE-Integrated AI Workflow

GitHub Copilot Agent was used directly inside Visual Studio Code during the final development and verification phase.

- Repository instructions: .github/copilot-instructions.md
- Four reusable prompts: .github/prompts/
- Four project context files: ai-context/
- Corrected zero-distance and zero-budget validation
- Added regression tests for verified defects
- Corrected Spring parameter binding through the Java -parameters option
- Generated the Trip lifecycle statechart
- Generated the corrected as-built UML class diagram
- Verified 29 automated tests with 0 failures
- Exported IDE chats and screenshots are preserved in ai-evidence/





Questions?

Thank you

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Plan clearly. Estimate early. Travel within budget.